

CURRICULUM VITAE

Giulio Gori, PhD

Full-time Researcher in Fluid Dynamics (Untenured), IIND-01/F,
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EDUCATION

- 28/1/2019 Ph.D. in Aerospace Engineering
Thesis: [*Non-Ideal Compressible Fluid-Dynamics: Developing a Combined Perspective on Modeling, Numerics and Experiments*](#). Department of Aerospace Science and Technology, Politecnico di Milano, Italy. Advisor: Prof. Alberto Matteo Attilio Guardone
- 22/7/2013 Master Degree in Aeronautical Engineering
Thesis: [*PoliMIce: un ambiente di simulazione per la previsione dell'accrescimento di ghiaccio su velivoli*](#). Department of Aerospace Science and Technology, Politecnico di Milano, Italy. Advisor: Prof. Alberto Matteo Attilio Guardone
- 22/9/2010 Bachelor's Degree in Aerospace Engineering
Department of Aerospace Science and Technology, Politecnico di Milano, Italy

CURRENT AND PREVIOUS POSITIONS

- 04/10/2025 –
Full-time Researcher in Fluid Dynamics (RTT), IIND-01/F, Department of Aerospace Science and Technology, Politecnico di Milano, Italy.
- 04/10/2021 – 03/10/2025
Full-time Researcher in Fluid Dynamics (RTDa), IIND-01/F, Department of Aerospace Science and Technology, Politecnico di Milano, Italy. [UN-BIASED](#) HORIZON-MSCA-PF-01
- 15/01/2021 – 03/10/2021
Post-doc researcher, CS2 H2020 [MONNALISA](#) Project, Department of Aerospace Science and Technology, Politecnico di Milano, Italy
- 02/10/2017 – 31/12/2020
[UTOPIAE](#) MSCA-ITN ESR, [Platon Team](#), INRIA/CMAP, École Polytechnique, France
- 16/06/2014 – 15/07/2017
Research fellow, [CREALab](#)/Department of Aerospace Science and Technology, Politecnico di Milano, Italy

QUALIFICATIONS

- 03/02/2022 – 03/02/2031
National Scientific qualification as associate in the Italian higher education system, in the call 2021/2023 (Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of 09/A1 - Aeronautical and aerospace engineering and naval architecture.
- 2014 (2nd session) – Not expiring
State Professional Examination for the qualification as Industrial Engineer – Section A.

SCIENTIFIC ACHIEVEMENTS AND PEER RECOGNITION

I delivered 30+ scientific contributions in about 10 years. Among these, 16 are peer-reviewed journal articles (10 as the main author), 12 are peer-reviewed conference proceedings (7 as the main author), 3 chapters in a book (1 as the main author), and many non-revised conference proceedings.

Since 2014, yearly citations of my works have grown consistently, attracting +260 #C (n. of citations from SCOPUS, excluding self-citations). Publications testify to the eclecticism of my Academic output.

I deem my most relevant publications to be:

[Ref.1 (2015)] **G. Gori** et al., *PoliMIce: A simulation framework for three-dimensional ice accretion*, Applied Mathematics and Computation, <https://doi.org/10.1016/j.amc.2015.05.081> (#C 52)

[Ref.2 (2022)] **G. Gori**, et al., *Modeling in-flight ice accretion under uncertain conditions*. Journal of Aircraft, <https://doi.org/10.2514/1.C036545> (#C 12)

Concerning in-flight ice accretion. I am the creator, and I have been the leading developer (from 2012 to 2017) of the PoliMIce ice accretion code first presented in Ref.1. This output of mine made it a long path, also thanks to other contributors, leading PoliMIce to be employed in several European research projects e.g., [H2020 UTOPIAE MSCA-ITN](#), [H2020 NITROS MSCA-EJD](#), [H2020 ICE-GENESIS](#) by Airbus, [H2020 MONNALISA CS2-JU](#), and [HE-MSCA-TRACES JD-DN](#). In Ref.2, I expose a challenge inherent in the approximation of the full icing model using standard (linear) uncertainty quantification techniques and propose a nonlinear method of superior performance.

[Ref.3 (2017)] **G. Gori**, et al., *Non-Ideal compressible fluid effects in oblique shock waves*, J. Phys.: Conf. Ser. <https://iopscience.iop.org/article/10.1088/1742-6596/821/1/012003> (#C 12)

[Ref.4 (2015)] S. Vitale, **G. Gori**, M. Pini, A. Guardone, T. D. Economon, F. Palacios, J.J. Alonso and P. Colonna, *Extension of the SU2 open source CFD code to the simulation of turbulent flows of fluids modelled with complex thermophysical laws*, 22nd AIAA CFD Conference. <https://doi.org/10.2514/6.2015-2760> (#C 36)

[Ref.5 (2020)] **G. Gori**, et al., *Accuracy assessment of the Non-Ideal Computational Fluid Dynamics model for siloxane MDM from the open-source SU2 suite*, European Journal of Mechanics - B/Fluids, <https://doi.org/10.1016/j.euromechflu.2019.08.014> (#C 15)

Non-Ideal Compressible Fluid Dynamics (NICFD) was recently acknowledged by ERCOFTAC (Special Interest Group #49). In Ref.3 I demonstrated the theoretical admissibility of an unprecedented phenomenon i.e., the so-called *non-ideal oblique shock-waves*, which fostered prolific research activities by the diverse scope of the citing research. Ref.4 testifies to my contribution to developing computational tools for NICFD. In Ref.5, I presented the first-ever experimental assessment of a computational model for NICFD, supporting [SU2](#) in becoming the academic standard.

[Ref.6 (2022)] **G. Gori**, et al., *A confidence-based aerospace design approach robust to structural turbulence closure uncertainty*, Comput Fluids, <https://doi.org/10.1016/j.compfluid.2022.105614> (#C 3)

is a recent example of my original thinking and my capacity to build upon previous knowledge and combine different expertise to achieve new goals. I am exploring new optimization methods for aerospace applications accounting for the credibility of the computerized model during the design process.

I am exposed to many different Research fields, and I collaborate with worldwide recognized experts

Prof. A. Guardone	Full professor, Dept Aerospace Science & Technology, Politecnico di Milano Italy Collaboration on non-ideal compressible fluid dynamics and in-flight icing
Prof. P. Colonna	Full professor, Chair of Propulsion and Power at TU Delft, Netherlands Collaboration on non-ideal compressible fluid dynamics
Prof. J.J. Alonso	Full professor in Aeronautics & Astronautics at Stanford University, CA, USA Collaboration on computational fluid dynamics and software development
Prof. G. Iaccarino	Full professor Mechanical Engineering at Stanford University, CA, USA Collaboration on uncertainty quantification and robust optimization
Prof. T. Magin	Professor, Aeronautics and Aerospace Department, Von Karman Institute, Belgium. Collaboration on aerothermodynamics and uncertainty quantification
Prof. P. Congedo	Head of Platon Team CMAP/INRIA, École Polytechnique, France Collaboration on uncertainty quantification and robust optimization
Prof. O. Le Maître	Research Director, Platon Team CNRS/CMAP/INRIA, École Polytechnique, France Collaboration on uncertainty quantification and robust optimization
Prof. M. Panesi	Director of the Center for Hypersonics and Entry Systems Studies (CHESS), University of Illinois Urbana-Champaign, USA, Collaboration on hypersonic flows and applications

Politecnico di Milano ranks 7th in QS World University Rankings by Subject 2023: Mechanical, Aeronautical & Manufacturing Engineering and entrusted me with the full responsibility of teaching the course “Fundamentals of Hypersonic Flows” at the Master of Science level, for which I have full responsibility. I began covering responsibility roles i.e., adjoint member of the PhD School Committee and Examination Board Member in the MSc Graduation Committee. I supervised +20 MSc students and I am co-supervising PhD ESR9 from the TRACES JD-DN.

In 2023 I was awarded a 197 k€ grant from the Italian government in the frame of the PRIN2022 call and I am the PI of the HERMES project. My proposal scored 91/100 and was selected among the +670 competitors.

In 2021, I wrote a successful proposal for the highly competitive HORIZON-MSCA-PF-01 fellowship call (UN-BIASED project, score 96.6%) obtaining a 173k funding and attracting the interest of newspaper “[la Repubblica](#)”.

In 2018, I was PI of a one-month expedition (3 people) to the Summer Program at the Center for Turbulence Research at Stanford University, with a project on Non-Ideal Compressible Fluid Dynamics concerning the robust optimization of turbine cascades. A small budget of about 3500 \$ was funded by Stanford University.

Since 2014, I have been a permanent member of the SU2 developers’ community, contributing to the early development of the SU2 NICFD and rotorcraft modules. In 2023, I was appointed by the SU2 Foundation “Coordinator of the local organizing committee” of the 4th and 5th Annual SU2 Conference, held in Varenna, Italy. Since 2024, I am a member of the SU2 Foundation Executive Board. Beginning 2026, I was appointed Secretary of the SU2 Foundation Board.

Concerning in-flight ice accretion, my scientific maturity was acknowledged with the invitation to contribute to the Handbook of Numerical Simulations of In-Flight Icing (2023) edited by Dr. W.G. Habashi. My contribution consists of authoring the chapter [Numerical simulations of in-flight icing under uncertain conditions](#) and co-authoring the chapter [Numerical Optimization of Electrothermal Ice Protection Systems](#).

I recently started a collaboration with Prof. Panesi, Director of the Center for Hypersonics and Entry Systems Studies (CHESS) at University of Urbana-Champaign Illinois. We exchange and co-supervise MSc students (7 MSc theses concluded) and collaborate on scientific aspects.

Since 2023, I was promoted Team Member of GARTEUR - Action Group RC/AG-26 - Noise Radiation and Propagation for Multirotor System Configurations. In 2023, I was selected as a young representative to present the “GARTEUR high level plan on future GoR work” at the [event](#) celebrating the 50th anniversary GARTEUR group.

PARTICIPATION TO NATIONAL AND INTERNATIONAL RESEARCH PROJECTS

- 2023 – [PRIN2022 HERMES](#) Role: PI and coordinator of the two national research units (PoliMI and PoliBA)
- 2023 – [HE TRACES MSCA-JD-DN](#) Role: co-supervisor of DR9
- 2023 – [GARTEUR \(GoR-RC\)](#) Role: unit member Action Group RC/AG-26
- 2022 – 2024 [HE UN-BIASED MSCA-PF](#) Role: PI
- 2021 – 2022 [H2020 MONNALISA CS2-JU](#) Role: unit member WP5
- 2021 – 2022 [H2020 ICE-GENESIS](#) Role: unit member WP10
- 2018 – 2018 CTR Summer program Role: coordinator of the research unit (3 people)
- 2017 – 2020 [H2020 UTOPIAE MSCA-ETN](#) Role: ESR3
- 2014 – 2017 [H2020 NSHOCK ERC](#) Role: Research Associate and PhD

FUNDINGS RECEIVED

- 2023 PRIN22-HERMES (197 k€)
- 2022 HORIZON-MSCA-PF-01 UN-BIASED (173 k€)
- 2018 CTR Summer Program at Stanford University (3,550 \$).

PUBLICATION RECORD

Peer Reviewed Journals

- 2025 **G. Gori**, Comput Fluids, <https://doi.org/10.1016/j.compfluid.2025.106875>
- 2025 **G. Gori** et al., J. Aircr, <https://doi.org/10.2514/1.C037765>
- 2025 P. Yan et al., Comput Phys Commun, <https://doi.org/10.1016/j.cpc.2024.109394>
- 2023 M. Gallia et al., J Aircr, <https://doi.org/10.2514/1.C037223>
- 2023 P. Yan et al., J Comput Appl Math, <https://doi.org/10.1016/j.cam.2023.115169>
- 2022 **G. Gori** et al., Comput Fluids, <https://doi.org/10.1016/j.compfluid.2022.105614>
- 2021 **G. Gori** et al., J Aircr, <https://doi.org/10.2514/1.C036545>
- 2021 **G. Gori** et al., Comput Fluids, <https://doi.org/10.1016/j.compfluid.2021.105081>
- 2020 **G. Gori** et al., Comput Fluids, <https://doi.org/10.1016/j.compfluid.2020.104550>
- 2020 N.Razaaly, et al, Appl Math Model, <https://doi.org/10.1016/j.apm.2020.01.048>
- 2020 **G. Gori** et al, Eur J Mech B <https://doi.org/10.1016/j.euromechflu.2019.08.014>
- 2018 **G. Gori**, et al., Appl. Math. Comput., <https://doi.org/10.1016/j.amc.2017.07.041>
- 2018 D. Vimercati, et al., J Fluid Mech, <https://doi.org/10.1017/jfm.2018.328>
- 2018 **G. Gori**, et al., J Aircr, <https://doi.org/10.2514/1.C034412>
- 2017 M. Zocca, et al., J Aircr, <https://doi.org/10.2514/1.C033750>
- 2015 **G. Gori** et al., Appl Math Comput, <https://doi.org/10.1016/j.amc.2015.05.081>

Chapter in Books

- 2023 **G. Gori** et al., Springer, Cham., https://doi.org/10.1007/978-3-030-64725-4_31-1
- 2023 M. Gallia et al., Springer, Cham., https://doi.org/10.1007/978-3-030-64725-4_37-1
- 2021 J. Reis et al., Springer, Cham., https://doi.org/10.1007/978-3-030-60166-9_1

Peer-Reviewed Conference Proceedings

- 2023 **G. Gori**, Spring Cham., https://doi.org/10.1007/978-3-031-30936-6_8
- 2020 **G. Gori** et al., https://doi.org/10.1007/978-3-030-80542-5_25
- 2019 **G. Gori** et al., [ORC2019](#), Athens, Greece
- 2019 N. Razaaly et al., [GPPS2019 Conference](#), Zurich, Switzerland
- 2018 N. Razaaly et al., [CTR Summer Program](#), Stanford University, USA
- 2017 **G. Gori** et al., *Energy Procedia*, <https://doi.org/10.1016/j.egypro.2017.09.151>
- 2017 D. Vimercati et al., *Energy Procedia*, <https://doi.org/10.1016/j.egypro.2017.09.231>
- 2017 P. Molesini et al. *Energy Procedia*, <https://doi.org/10.1016/j.egypro.2017.09.152>
- 2017 M. Pini et al., *J. Phys.: Conf. Ser.* <https://iopscience.iop.org/article/10.1088/1742-6596/821/1/012013>
- 2017 **G. Gori** et al., *J. Phys.: Conf. Ser.* <https://iopscience.iop.org/article/10.1088/1742-6596/821/1/012005>
- 2017 **G. Gori** et al., *J. Phys.: Conf. Ser.* <https://iopscience.iop.org/article/10.1088/1742-6596/821/1/012003>
- 2015 **G. Gori** et al., [ASME ORC2015](#), Brussels, Belgium

Conference Proceedings

- 2025 L. Abergo et al., *AIAA SCITECH 2025 Forum*. Orlando (USA). <http://hdl.handle.net/11311/1281748>
- 2025 F. Vegetti et al., *ERF2025*, Venezia, Italy. <http://hdl.handle.net/11311/1300593>
- 2024 **G. Gori** et al, *ICAS2024*, Firenze, Italy. <http://hdl.handle.net/11311/1273347>
- 2024 **G. Gori**, *ICCFD12*, Kobe Japan. <http://hdl.handle.net/11311/1270565>
- 2024 A. Meo et al., *AIAA AVIATION 2024 Forum*. Las Vegas (USA), <https://doi.org/10.2514/6.2024-3563>
- 2024 A. Perlini et al, *AIAA AVIATION 2024 Forum*. Las Vegas (USA), <https://doi.org/10.2514/6.2024-3753>
- 2024 E. Monti et al., *AIAA SCITECH 2024 Forum*. Orlando (USA) <https://doi.org/10.2514/6.2024-0654>
- 2024 C. Capecchi et al., *AIAA SCITECH 2043 Forum*. Orlando (USA) <https://doi.org/10.2514/6.2024-0655>
- 2023 **G. Gori** et al., *AIAA AVIATION 2023 Forum*. San Diego (USA) <https://doi.org/10.2514/6.2023-4422>
- 2023 F. Caccia et al., *AIAA AVIATION 2023 Forum*. San Diego (USA) <https://doi.org/10.2514/6.2023-3221>
- 2022 **G. Gori** et al., *AIAA AVIATION 2022 Forum*. Chicago (USA) <https://doi.org/10.2514/6.2022-3532>
- 2022 A. Rausa et al., *AIAA AVIATION 2022 Forum*. Chicago (USA) <https://doi.org/10.2514/6.2022-3902>.

- 2022 F. Auteri et al., *AIAA AVIATION 2022 Forum*. Chicago (USA) <https://doi.org/10.2514/6.2022-4149>
- 2021 **G. Gori** and A. Guardone, *XXVI AIDAA Congress*, Pisa, Italy.
- 2021 **G. Gori**, et al., *AIAA AVIATION 2021 Forum*. (Virtual event) <https://doi.org/10.2514/6.2021-2683>
- 2021 T. Bellosta et al., *AIAA AVIATION 2021 Forum*. (Virtual event) <https://doi.org/10.2514/6.2021-2645>
- 2019 **G. Gori** et al., *FAR2019*, Monopoli (BA), Italy.
- 2019 B. Arizmendi et al., *AIAA AVIATION 2019 Forum*. Dallas (USA) <https://doi.org/10.2514/6.2019-3464>
- 2018 **G. Gori** et al., ISSW31, Nagoya (JPN). https://doi.org/10.1007/978-3-319-91017-8_93
- 2018 M. Zocca et al., *ECFD7*, Glasgow, United Kingdom.
- 2018 N. Razaaly et al., *ECFD7*, Glasgow, United Kingdom.
- 2015 S. Vitale et al., *AIAA AVIATION 2015 Forum*, Dallas (USA) <https://doi.org/10.2514/6.2015-2760>
- 2015 **G. Gori** et al., *AIAA AVIATION 2015 Forum*, Dallas (USA) <https://doi.org/10.2514/6.2015-3019>

MOBILITY

- 2019 Von Karman Institute for Fluid Dynamics, Belgium (4 mths). Uncertainty Quantification for hypersonic flows and heat shield ablation for atmospheric entry applications.
- 2018 Center for Turbulence Research at Stanford University (1.5 mths), Palo Alto, CA, USA. Development of robust optimization approaches for Organic Rankine Cycle applications.
- 2016 UT Twente, Faculty of Engineering Technology, Enschede, Netherlands (3 mths). Developing numerical methods for sliding mesh interfaces in computational fluid dynamics.
- 2014 Aerospace Design Lab (ADL) at Stanford University (1 mths), Palo Alto, CA, USA. Development and implementation of the SU2 non-ideal compressible-fluid dynamics solver.

AWARDED FELLOWSHIPS

- 2022 Post-doctoral Fellowship HORIZON-MSCA-PF-01, UN-BIASED “UNcertainty quantification and modelling Bias Inhibition by means of an Agnostic Synergistic Exploitation of multi-fidelity Data”, Department of Aerospace Science and Technology, Politecnico di Milano, Italy
- 2021 Temporary Research Fellowship for Research Activities UOR DAER “Development of simplified models for the aerodynamics of wings at high angle of attack” CS2-H2020 MONNALISA/Department of Aerospace Science and Technology, Politecnico di Milano, Italy
- 2017 – 2020 Early-Stage Researcher Fellowship, H2020-MSCA-ITN-2016, UTOPIAE-ESR3 “Inference and Design of Experiments in Large Scale Flow Problems”, INRIA/Centre de Mathématiques Appliquées, École Polytechnique, IPP, France
- 2015 – 2017 Temporary Research Fellowship for Research Activities UOR DAER “Metodi numerici per la simulazione di correnti di fluidi comprimibili non-ideali” CREALab/Department of Aerospace Science and Technology, Politecnico di Milano, Italy
- 2014 – 2015 Temporary Research Fellowship for Research Activities UOR DAER “Simulazione numerica di correnti di gas densi con codici fluidodinamici per griglie chimera” CREALab/Department of Aerospace Science and Technology, Politecnico di Milano, Italy

INSTITUTIONAL RESPONSABILITIES

At the Department of Aerospace Science & Technology, Politecnico di Milano, Italy:

- 2024 – School Delegate in charge of the Admission to the Master of Science programmes
- 2023 – Adjunct member, PhD School Committee
- 2022 – Member, Learning Program Council
- 2022 – Member of the Communication and Event Organization Group, social media manager
- 2022 – Member of the Master of Science Graduation Committee (Examination Board Member)
- 2022 – Coordinator of the Special Interest Group for Machine Learning in Aerospace Applications
- 2014 – 2017 Coordinator of the SU2 open-source CFD solver User Group

TEACHING ACTIVITY

At the Department of Aerospace Science & Technology, Politecnico di Milano, Italy:

2024 –	Co-lecturer – “Aerothermodynamics” (Partial responsibility for lectures and exams)
2023 –	Lecturer – “Fundamentals of Hypersonic Flows” (Full responsibility for lectures and exams)
2022 – 2022	Assistant – “Fundamentals of Hypersonic Flows” (Students’ project tutoring)
2021 – 2022	Assistant – “Compressible Fluid Dynamics” (Exercise sessions and numerical labs)
2021 – 2022	Assistant – “Computational Fluid Dynamics” (Students’ project tutoring)

SUPERVISING ACTIVITY

2024 –	PhD Advisor of Mr. Alberto Perlini
2023 –	Supervisor of 2 Research Fellows (HERMES Project)
2023 –	PhD co-supervisor of TRACES-ESR9 Mr. Omar Kahol
2014 –	Supervisor/co-supervisor of +30 (including ongoing) MSc students developing their final thesis

ORGANIZATION OF SCIENTIFIC MEETINGS

2026	Member, AIDAA Technical Committee, Hypersonics 2026 , Napoli, Italy
2026	Coordinator of the Local Organizing Committee, 7 th SU2 Conference, Varenna, Italy
2025	Coordinator of the Local Organizing Committee, 6th SU2 Conference , Varenna, Italy
2024	Coordinator of the Local Organizing Committee, 5th SU2 Conference , Varenna, Italy
2023	Coordinator of the Local Organizing Committee, 4th SU2 Conference , Varenna, Italy
2022	Organizer of symposium “ <i>Numerical Simulation of Ice Accretion</i> ”, ESCO2022 , Plizen, CZ
2019	Member of Organizing Committee (Local Organizer), UQOP2019 , Paris, France
2017	Staff member, 4th International Seminar on ORC Power Systems , Milano, Italy
2016	Staff member, NICFD-PP 2016 , Varenna, Italy

MEMBERSHIP OF SCIENTIFIC SOCIETIES, VOLUNTEERING, AND REVIEWING ACTIVITIES

2026	Secretary of the SU2 Foundation Executive Board
2024	Member of the SU2 Foundation Executive Board
2023	Team Member GARTEUR - Action Group RC/AG-26 - Noise Radiation and Propagation for Multirotor System Configurations
2021 –	Member of the Italian Association for Aeronautics and Astronautics (AIDAA)
2015 –	Scholarly peer reviewer
2014 –	Volunteer Member of the SU2 Principal Developers Team

SOFTWARE DEVELOPMENT

PoliMice	Core developer member of the integrated CFD-multiphase PoliMice software (2012-2017).
SU2	Member of the Principal Developers team of the SU2 open-source CFD solver (since 2014).
PoliUQ	An in-house Python library for Uncertainty Quantification and robust optimization.

Milano,
Jan 8th, 2026



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